

Food Delivery Analysis Dashboard — Beginner's Build Guide

A complete, from-scratch walkthrough for building the project described in your PDF, using your actual `Dataset.csv` (800 orders, 11 columns). Written for someone who has never opened Power BI before.

0. What you're building, in plain terms

You have one flat table of orders. You're going to:

1. Load it into Power BI and clean it up (Power Query)
2. Add a couple of helper columns and a proper Date table
3. Write ~10 DAX measures
4. Build 3 report pages: **Overview, Delivery Analysis, Restaurant Analysis**

That's it. No complex relationships needed since it's a single table — this is actually the *easiest* kind of Power BI project to learn on, which makes it a good first project.

1. Install & open Power BI Desktop

1. Search "Power BI Desktop" in the Microsoft Store (Windows) — it's free. Mac users need a Windows VM or Power BI Service (browser) instead, since Desktop is Windows-only.
 2. Open it, close the startup splash screen.
 3. Home tab → **Get Data** → **Text/CSV** → select your `Dataset.csv`.
 4. A preview window pops up. Don't click Load yet — click **Transform Data** instead. This opens Power Query Editor, where all cleaning happens *before* the data lands in your report.
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2. Clean the data in Power Query

This is the step most beginners skip, and it's the one that separates a "student project" from a "portfolio project." Here's exactly what to do, in order.

2.1 Fix the data types

Power BI often guesses column types wrong on import. Click each column header's little type icon and set:

- (Order ID), (Customer ID), (Restaurant Name), (Food Category), (Payment Method), (City), (Order Status) → **Text**
- (Order Value), (Delivery Time), (Rating) → **Decimal Number**
- (Order Date) → see below, don't just click "Date" yet

2.2 Fix the date column carefully

Your dates are in **DD-MM-YYYY** format (e.g. (24-11-2025), (13-12-2025)). If you just right-click → Change Type → Date, Power BI's default US locale will try MM-DD-YYYY and either scramble your dates silently or error out on any day above 12.

Correct way:

1. Right-click the (Order Date) column header → **Change Type** → **Using Locale...**
2. Data Type: **Date**. Locale: **English (United Kingdom)**.
3. Click OK. Now it parses correctly as day-month-year.

2.3 Understand the nulls before you touch them

Check this first (Power Query shows null counts if you click a column and look at the bottom status bar profiling, or View → Column distribution):

Column	Nulls	Why
(Delivery Time)	124	Every one of these is a (Pending) order — hasn't been delivered yet, so there's no time to record
(Rating)	236	These are the 124 (Pending) + 112 (Cancelled) orders — you can't rate an order you never received

Don't fill these with 0 or an average. A 0-minute delivery or a 0-star rating would badly distort your averages. The correct move is to just leave them as blanks (nulls) — DAX's (AVERAGE()) automatically ignores blanks, which is exactly the behavior you want. This is worth a line in your project write-up: it shows you understood the data instead of blindly imputing it.

2.4 Trim and clean text

Select all text columns → Transform tab → **Format** → **Trim** (removes stray spaces) and **Clean** (removes non-printable characters). Cheap insurance against duplicate categories like ("Pizza ") vs ("Pizza").

2.5 Add a couple of helper columns

Still in Power Query, Add Column → Custom Column:

- **Revenue Bucket** (useful for a distribution visual later):

```
if [Order Value] < 500 then "Low (<₹500)"  
else if [Order Value] < 1000 then "Medium (₹500–1000)"  
else "High (>₹1000)"
```

- **Delivery SLA Flag** (only meaningful for Delivered orders):

```
if [Order Status] <> "Delivered" then "N/A"  
else if [Delivery Time] <= 30 then "On Time"  
else "Delayed"
```

(Adjust the 30-minute threshold if you want a different SLA target — this is a judgment call you get to make and justify.)

Click **Close & Apply** on the Home tab. Your cleaned table now loads into the model.

3. Build a proper Date table

Even with one data table, a separate Date table is standard practice and lets you do month/quarter slicers, growth %, etc. correctly.

Home tab → **New Table**, then paste:

```
DateTable = CALENDAR(MIN(Dataset[Order Date]), MAX(Dataset[Order Date]))
```

Then add columns to it (New Column, run these one at a time):

```
Year = YEAR(DateTable[Date])  
MonthNumber = MONTH(DateTable[Date])  
MonthName = FORMAT(DateTable[Date], "MMM")
```

Go to **Model view** (left sidebar icon) and drag `DateTable[Date]` onto `Dataset[Order Date]` to create a relationship (one-to-many, DateTable → Dataset). This is the standard star-schema pattern — one small lesson here goes a long way if you ever work with multiple fact tables later.

4. DAX Measures

Home tab → **New Measure**, create each of these in a dedicated "Measures" table if you want to stay organized (optional but tidy — New Table → `Measures = {1}`), then create all measures there instead of scattering them across Dataset).

DAX

```
Total Orders = COUNTROWS(Dataset)
```

```
Delivered Orders = CALCULATE([Total Orders], Dataset[Order Status] = "Delivered")
```

```
Cancelled Orders = CALCULATE([Total Orders], Dataset[Order Status] = "Cancelled")
```

```
Cancellation Rate % = DIVIDE([Cancelled Orders], [Total Orders])
```

```
Total Revenue = SUM(Dataset[Order Value])
```

```
Delivered Revenue = CALCULATE([Total Revenue], Dataset[Order Status] = "Delivered")
```

```
Avg Delivery Time = AVERAGE(Dataset[Delivery Time])
```

```
Avg Rating = AVERAGE(Dataset[Rating])
```

```
On Time Delivery % =
```

```
DIVIDE(
```

```
    CALCULATE([Total Orders], Dataset[Delivery SLA Flag] = "On Time"),
```

```
    CALCULATE([Total Orders], Dataset[Order Status] = "Delivered")
```

```
)
```

```
Revenue Growth % =
```

```
VAR CurrentMonthRevenue = [Total Revenue]
```

```
VAR PreviousMonthRevenue =
```

```
    CALCULATE([Total Revenue], DATEADD(DateTable[Date], -1, MONTH))
```

```
RETURN
```

```
    DIVIDE(CurrentMonthRevenue - PreviousMonthRevenue, PreviousMonthRevenue)
```

Why `Delivered Revenue` matters: your Order Value column presumably charges the customer regardless of status, but "revenue" in a real business sense often only counts fulfilled orders. Having both `Total Revenue` and `Delivered Revenue` as separate measures lets you show you understand that distinction — mention it in your dashboard notes.

5. Dashboard Pages

Page 1 — Overview

- **Three Card** visuals across the top: `Total Orders`, `Total Revenue` (or `Delivered Revenue`), `Avg Rating`
- A **Donut chart**: `Order Status` by `Total Orders` (shows Delivered/Pending/Cancelled split at a glance)
- A **Line chart**: `DateTable[MonthName]` on X-axis, `Total Revenue` on Y-axis — your revenue trend
- A **Bar chart**: `Food Category` by `Total Orders` — popular categories
- A **Slicer** for `City` and one for `Order Status` at the top, so every visual on the page is filterable

Page 2 — Delivery Analysis

- **Clustered bar chart**: `City` (Y-axis) by `Avg Delivery Time` (X-axis) — instantly shows which cities are slow
- **Stacked bar chart**: `Order Status` by `Total Orders`, broken down further by `City` — this is your "Order Status Analysis" from the brief
- **Card**: `On Time Delivery %`
- **Scatter chart** (nice touch, shows real analysis): `Avg Delivery Time` vs `Avg Rating` by `City` — tests the hypothesis that slower delivery hurts ratings
- Filter this whole page to `Order Status = Delivered` (page-level filter) since delivery time/rating only exist for those rows

Page 3 — Restaurant Analysis

- **Table or matrix**: `Restaurant Name`, `Total Orders`, `Total Revenue`, `Avg Rating` — sort by Total Revenue descending
 - **Top N bar chart**: Top 10 restaurants by `Total Revenue` (use a Top N filter on the visual)
 - **Card**: highest-rated restaurant (use a Top N=1 table or a measure with `(TOPN)`)
 - **Treemap**: `Food Category` by `Total Revenue` — shows which categories drive the most money, not just the most orders (a category can have lots of cheap orders but low revenue, or vice versa — worth calling out as an insight)
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6. Design polish (quick wins)

- Format → Page background: solid dark or light neutral, consistent across all 3 pages
 - Use one accent color for "good" (Delivered/On Time) and one warning color for "Cancelled/Delayed" — consistency across pages makes it look intentional
 - Add a text box title on each page + a small "as of [date]" footer
 - Turn off default visual titles' auto-generated text if it looks cluttered; write your own short titles instead ("Where are we losing time?" reads better than "Avg Delivery Time by City")
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7. What to say in your project write-up / resume bullet

Since this is a portfolio piece, mention the reasoning, not just the output:

- You identified that missing (Delivery Time)/(Rating) values were structurally explained by (Order Status), not data errors, and handled them accordingly instead of imputing
- You separated (Total Revenue) from (Delivered Revenue) to distinguish gross demand from fulfilled revenue
- You built a proper Date dimension table instead of relying on the raw date column, enabling correct month-over-month growth calculations

That kind of explanation is what turns "I followed a tutorial" into "I understood the problem."

8. Suggested order to actually do this in

1. Load + clean data (Section 2) — ~20 min
2. Date table + relationship (Section 3) — ~10 min
3. All DAX measures (Section 4) — ~20 min, expect a few typos, that's normal
4. Overview page — ~20 min
5. Delivery Analysis page — ~20 min
6. Restaurant Analysis page — ~20 min
7. Formatting pass across all pages — ~15 min

Roughly a 2-hour project spread over one or two sessions. If you get stuck on a specific DAX error or a visual not showing what you expect, describe exactly what you see and I'll help you debug it.